

SOLID STATE SHOPPING

Agent001 finds the best SSDs for various use-case scenarios

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Solid State Drives, or SSDs, have been around for quite a while. The only reason they're not selling as high in volumes is because of the high cost per GB. Considering we're a price-conscious market, it's no surprise that I hear things like "Why should I spend 10,000 bucks on a 128 GB SSD when I can get a 1 TB HDD at that price?"

The problem lies with the perception that an SSD is just a speedier alternative to the humble HDD and when you go with that mindset to purchase an SSD, the pricing will obviously draw flak. So let's set the record straight – an SSD is primarily meant for faster access of data, and yes, it can store data, but it can't match the cost per GB offered by mechanical drives. So think of an SSD as an addition to your traditional HDD, unless you have money to burn.

An SSD uses NAND-based flash memory which retains data even without power and that's how it's distinguished from the conventional HDD. An SSD uses microchips that retain data in non-volatile memory chips containing no moving parts. Most importantly, the data flow is controlled by a central controller chip, and naturally different controller chips offer vastly different performances. Initially, there were only two controller manufacturers namely Intel and Indilinx but there are many more now – Jmicron, Toshiba, Marvell, SandForce and Samsung among others.

Team Digit members wanted to upgrade their systems, so keeping their requirements in mind, I headed out to Lamington Road to enquire about SSDs.

Team member: Nimish

Requirement: Boot drive

Budget: Around ₹5,000

Nimish doesn't want to spend much considering he has a mid-range system at home. So he's only looking for an SSD that

will have his Windows installation and some essential utilities. For such usage, the maximum limit should be around 64 GB just to have enough space if you want to add your Office and security suites. Kingston SSDNow V100 64 GB drive offers good value for money at ₹5,500. The other drive I came across was Corsair Nova Series 40 GB SSD at ₹4,500. While the Kingston drive houses the Toshiba JM618 controller, the Corsair one uses an Indilinx controller. Both these drives support TRIM, which is very important for any SSD purchase, as it doesn't affect the write speeds once all the blocks are full of data and data needs to be written over a block.



Kingston HyperX 240GB

Team member: Jayesh

Requirement: Boot drive for his games

Budget: Around ₹12,000

Jayesh is a gamer and can't wait for the slow loading times offered by his 5400 RPM drive to end. He wants to invest in an SSD that will double up as his boot drive and hold some of his favourite heavy games. So the capacity I was looking at was around 128 GB, as that will give him enough space for the OS, the essential utilities and his game collection. Corsair Force GT 120GB at ₹10,000 is a good bet and it also won our Best Performer award in October 2011. It has the fast SandForce SF2281 controller which not only gives great read speeds, but also great sequential write speeds. And it has a SATA 6 Gbps interface. Its competitor, the Kingston



SSDNow V+ 100 was selling for around ₹11,000. This is also a great SSD, but has a comparatively slower Toshiba controller. OCZ Agility 3 120GB drive at ₹11,200 is another SATA 3 interface drive with the SandForce SF2281 controller, so great performance is expected. Game loading times with these drives is much more speedier than on a regular HDD.

Team Member: Robert

Requirement: Single performance drive

Budget: Around ₹30,000

Robert is a power user and despite using Velociraptors in RAID 0, he still feels that his system is slow. He willing to forego terabyte capacities, as long as he gets a blazingly fast SSD. Since he uses a lot of heavy applications such as Adobe's Creative Suite and also games in his free time, the drive needs decent capacity.

Kingston's HyperX 240GB SSD is the first thing I looked for, as that was our Zero1 Award winner last year. One of the fastest and highest capacity SSDs we tested in our labs. Corsair Force GT 240 GB drive, the higher capacity version of Corsair Force GT 120GB was selling for ₹29,000. The other drive we came across was OCZ Vertex 3 SATA 6 Gbps drive which was slightly over our budget at ₹33,000.

So those were the drives I found were good enough for the respective use case scenarios. But you may have noticed Intel's Z68 boards advertising about the Intel Smart Response Technology, which allows you to use an SSD in a RAID combination with your HDD such that your SSD can be used as cache memory, enabling quick access of most frequently used applications. Since there's a cap of 64 GB for Intel SRT, we would suggest going for any of the drives if you belong to the first category. **d**